

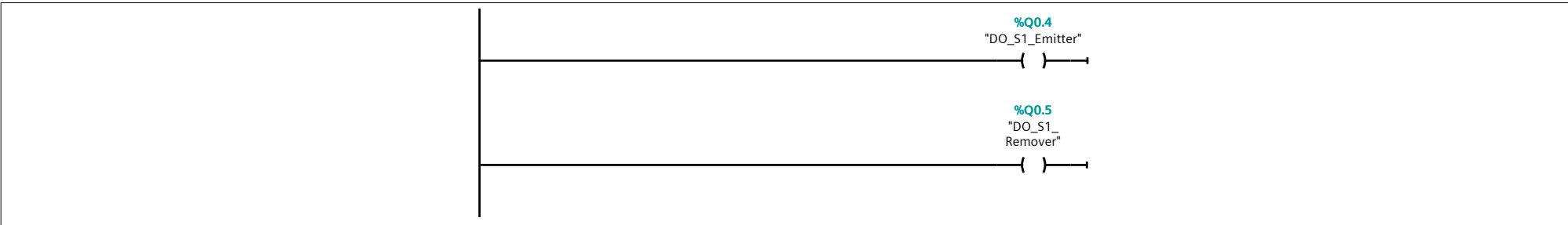
Station1 [FC1]

Station1 Properties							
General							
Name	Station1	Number	1	Type	FC	Language	LAD
Numbering	Automatic						
Information							
Title	Infeed & Color Sorting	Author	Youness Tebbaq	Comment	"Station 1 (Infeed & Col-or Sorting). The emitter spawns (Blue/Green) lids on-to the main conveyor. A dif-fuse sensor starts the 4 m belt only when the 2 m belts are free (anti-jam). The vi-sion sensor triggers a push-er request whene a blue lid detected; a short timer then fires the pusher. When the pusher reaches its front lim-it it retracts and the request flag is cleared. The station runs in a continuous loop with no start/stop logic."	Family	
Version	0.1	User-defined ID					

Name	Data type	Default value	Comment
Input			
Output			
InOut			
Temp			
Constant			
▼ Return			
Station1	Void		

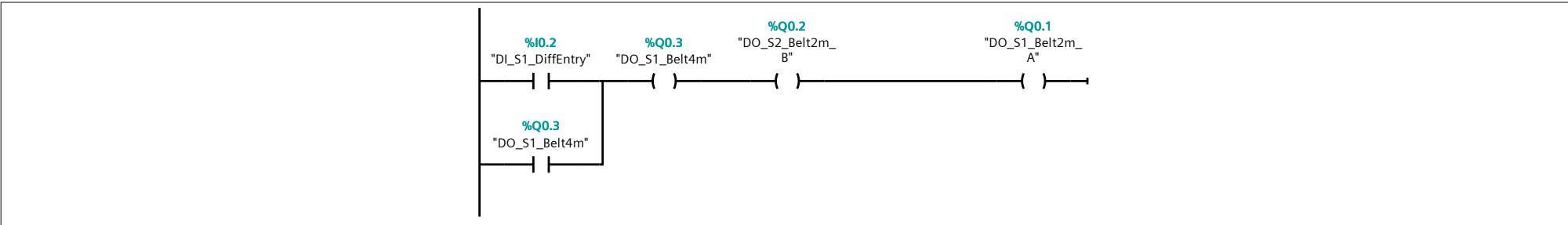
Network 1: Always run — Emitter and Remover

Emitter and Remover are permanently ON. The emitter continuously spawns Blue/Green lids into the line; the remover continuously deletes Blue lids at the end of the diverted lane .



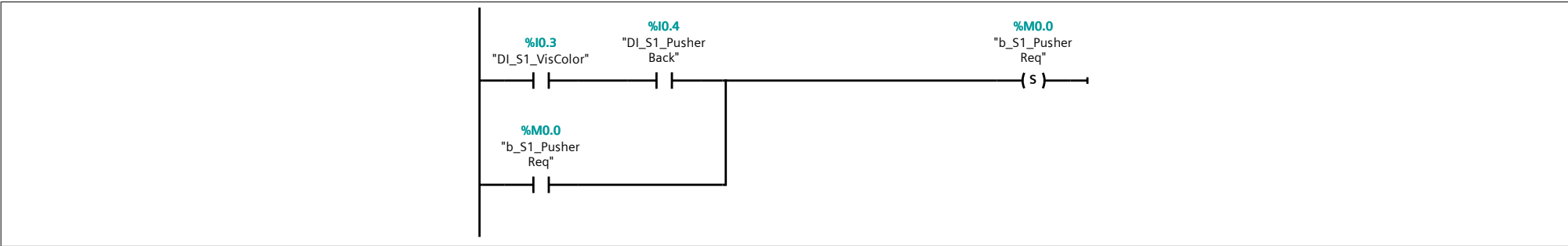
Network 2: Start the 4 m belt when an item is present

Enable the long 4 m belt (DO_S1_Belt4m) when:
- a part is detected by the diffuse entry sensor (DI_S1_DiffEntry), OR
- the belt is already running because a part is on it.



Network 3: Latch the pusher request when the vision sensor sees a part and the pusher is at home

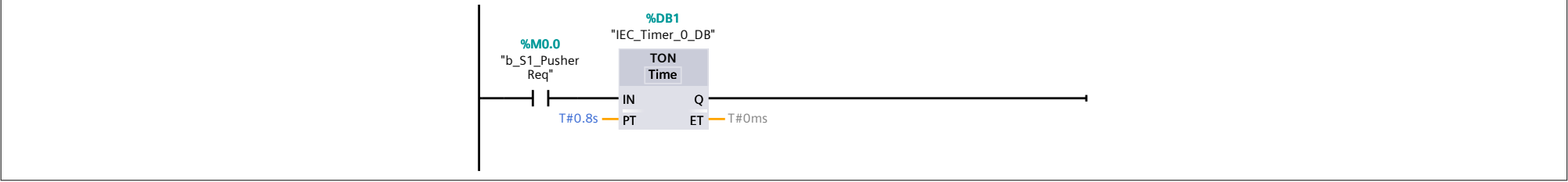
Latch the pusher request flag (b_S1_PusherReq) when:
- the vision sensor detects a part (DI_S1_VisColor), AND
- the pusher is fully retracted (DI_S1_PusherBack).
The flag stays set because the network ORs the flag back into itself, so it remains true even after the vision sensor has moved on.



Network 4: Short timer delay after the request is latched

Run a 0.8-second TON timer triggered by the pusher request flag. The small delay gives the item time to travel from the vision sensor to the pusher position before the pusher extends. If you

find the pusher fires too early or too late, change the PT (preset time) of this timer block.



Network 5: Extend the pusher when the timer expires

When the timer elapses, energise the pusher coil (DO_S1_Pusher) and latch it with the SET instruction. The pusher will stay extended until the front limit sensor resets it (Network 6).



Network 6: Retract the pusher when it reaches the front limit

When the pusher cylinder reaches its fully-extended position (DI_S1_PusherFront = TRUE), reset the pusher coil. The pusher retracts automatically (spring-return in Factory I/O).



Network 7: Clear the pusher request when the timer expires

When the timer elapses, also reset the request flag (b_S1_PusherReq). The station is now ready to receive the next part.

